SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING   COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Data Interpretation (Medium)
Weightage: 20%

QUESTIONS: Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any **incorrect or invalid fields** in the header.

Answer: The field Content-Length: -150 is incorrect and invalid because Content-Length cannot accept negative values.

2. Analyze the issue with **Content-Length** and explain its impact.

Answer: The Content-Length header informs the client of the message payload size in octets/bytes. A negative value is semantically invalid.

- Impact: The parser of the user agent (browser) or intermediary proxy servers will fail to determine the message boundaries. This causes socket connection hangs, dropped data packets, or renders the connection susceptible to HTTP Response Smuggling vulnerabilities.

3. Determine whether the **status code** matches the response correctness.

Answer: No, the status code 200 OK indicates that the request has succeeded and the payload is valid. However, since the response headers contain an invalid Content-Length field, the response is malformed. The server should have returned a server-side error code (such as 500 Internal Server Error) instead of claiming success.

4. Suggest **corrections** for the identified errors.

Answer: The Content-Length field should be updated to show the exact positive byte size of the payload. The corrected header is:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: 150
Connection: keep-alive
```

2: DOM Structure Analysis

Consider the following DOM tree representation:

```
<html>
  <head>
    <title>TestPage</title>
  </head>
  <body>
    <div>
      <p>HelloWorld
    </div>
  </body>
</html>
```

Questions:

1. Identify **missing or improperly closed elements**.

Answer: The <p> tag enclosing "HelloWorld" is unclosed. It must be closed with a corresponding </p> tag before the containing <div> element is closed.

2. Analyze **how the browser will interpret** this structure.

Answer: Modern browsers parse markup using the HTML5 parser specification rules:

- Parsing: When the parser encounters the closing </div> tag, it detects that a paragraph element (<p>) is still open. It implicitly auto-closes the open <p> tag to maintain correct XML/HTML nesting and then closes the <div> element.

3. Determine the **impact on rendering and layout**.

Answer: The impacts are:

- Visual Rendering: Because modern browsers autocorrect missing paragraph tags, the visual layout will likely display normally.

- CSS and Scripting issues: If stylesheets or scripts target strict element structures (e.g., div > p or sibling relations), parsing corrections might shift the logical tree structure in legacy browsers, leading to unexpected behavior.

4. Suggest **corrections** to make the DOM valid.

Answer: `Explicitly close the paragraph tag before closing the div container:`

```
<html>
  <head>
    <title>TestPage</title>
  </head>
  <body>
    <div>
      <p>HelloWorld</p>
    </div>
  </body>
</html>
```

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in **data transmission between GET and POST**.

Answer: - GET: Appends query parameters directly to the request URL. Data size is restricted by URL length limits, and parameters are fully exposed.

- POST: Transmits data in the body section of the HTTP request payload. It imposes no URL size limitations and keeps parameters out of the address bar.

2. Analyze which method is more **suitable for login** and why.

Answer: POST is the correct method for log-ins because POST parameters are transmitted in the HTTP request payload body, avoiding exposure of sensitive passwords in browser address bars, history logs, and proxies.

3. Identify **security risks** in using GET for authentication.

Answer: Using GET for authentication exposes user credentials to multiple vulnerabilities:

- URL History Leak: Browsers keep local logs of all visited URLs (including username/password query variables).

- Server Log Exposure: Web servers log full request URLs, storing plain-text passwords in system disk files.

- Referer Header Leak: External links clicked from the authenticated page will transmit query credentials to external third-party hosts via the HTTP Referer header.

4. Suggest **best practices** for secure login implementation.

Answer: Secure authentication guidelines include:

- Use HTTPS (TLS/SSL): Encrypt connection traffic so credentials cannot be read by sniffing network nodes.

- Use HTTP POST: Keep user details inside the HTTP body.

- Set Cookie Protections: Flag authorization cookies as HttpOnly, Secure, and SameSite.

- Implement Anti-CSRF protections on submit endpoints.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Sample</title>
  </head>
  <body>
    <h1>Welcome
    <p>This is a paragraph
    <img src="image.jpg">
  </body>
</html>
```

Questions:

1. Identify **missing closing tags**.

Answer: The missing closing tags are:

- The heading tag <h1> lacks a closing </h1> tag.

- The paragraph tag <p> lacks a closing </p> tag.

2. Analyze issues with the **<img>** tag.

Answer: The issues with the <img> tag are:

- Lack of alt attribute: Essential for accessibility compliance (W3C/WCAG) so screen readers can describe the image.

- No dimension attributes: Missing width and height attributes, which can cause Cumulative Layout Shift (CLS) during page load.

3. Determine **how browsers will handle** these errors.

Answer: Modern browsers correct these missing tags automatically during DOM construction:

- The open <h1> tag is closed when the parser hits the <p> element.

- The open <p> tag is closed when the parser hits the <img> tag.

- The missing alt attribute is ignored by the parser, but it fails standard W3C accessibility compliance checks.

4. Suggest a **corrected version** of the code.

Answer: The corrected W3C-compliant document is:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sample</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is a paragraph</p>
  <img src="image.jpg" alt="Sample Image" width="500" height="300">
</body>
</html>
```

5: Browser-Server Communication Flow

The following sequence describes a web request:

```
1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page
```

Questions:

1. Interpret **each step in the communication flow**.

Answer: Interpretation of the steps:

1. URL entry: User requests a address resource (e.g. typing or clicking).

2. DNS lookup: The system translates hostnames to numeric target IP addresses.

3. HTTP request: The browser initializes TCP (and TLS) handshakes and sends the request method/headers/body.

4. Server processing: Web server handles the endpoint route, runs controllers/databases, and outputs the reply.

5. Response return: The server streams back status code, response headers, and layout/page data.

6. Browser rendering: Browser engines construct DOM/CSSOM trees, compute layouts, and draw paint pixels.

2. Identify **where delays can occur**.

Answer: Latency delays can accumulate at several steps:

- Step 2 (DNS Lookup): High lookup times if domain namespaces are not cached.

- Step 3 (Network Latency): Slow packet travel during TCP handshake and TLS security handshakes.

- Step 4 (Server Logic): Slow server responses due to bad database queries or high CPU workloads.

- Step 6 (Layout blocking): Large render-blocking JS files or poorly placed stylesheets.

3. Analyze the role of **DNS** in the process.

Answer: The Domain Name System (DNS) acts as the telephone directory for the web. Since routers and network protocols communicate via IP addresses (like 192.0.2.1), DNS resolves human-friendly names (like example.com) to numeric IP coordinates.

4. Suggest **optimizations** to improve performance.

Answer: Optimizations to minimize web request times:

- Use DNS Prefetching: Resolve third-party host domains before they are clicked.

- Implement CDNs: Place static content caches geographically closer to users.

- Enable Connection Reuse: Maintain persistent TCP connections using Connection: keep-alive.

- Compress Assets: Optimize page elements using gzip or Brotli formats.

Code of Conduct:

I ___Enamala Gowtham (192311190) _________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Enamala Gowtham

RUBRICS FOR CALCULATION:

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand